FLOOD PEAK ESTIMATION OF SUB-BASINS OF DELHI NCT

Project Thesis

Submitted by

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ABSTRACT

Urban flooding is the buildup of water in a region caused by excessive rainfall surpassing the existing ability of drainage systems or huge water spills from bodies of water that exceed normal limitations. It may have widespread effect, affecting a noticeably big region, or it could be localized affecting a small area. Since the last 20 years the phenomena of urban floods increases in Delhi due to the rapid urbanization and sudden climate change. These floods cause a loss of life and a lot of harm to property in the city. Therefore estimation/ quantification of floods is an advanced way for engineered establishments in multiple aspects of development and progress.

This study was focussed on development of understanding on hydrological parameters, different techniques for flood estimation in phased manner. Estimation of floods for various sub-basins of Delhi NCT, Rational and Unit hydrographs approaches were used.

Collection of essential raw data like Probable Maximum Precipitation and its hourly variation for 1-day, Catchment area of sub-basins and Population etc. were collected from publications of Govt institutes like IMD, CWC and IIT Delhi. PMP atlas and catchment-based characteristics like average slope, Length of the drainage path etc. were extracted from the Maps developed using ArcGIS.

Considering the limitations of Rainfall data from IMD and usage of Unit hydrograph technique for smaller basins, Rational approach was adopted for Aruna Nagar and Old Chanderwal basins (both having Catchment area less than 50 km^2), eventually flood of recurrence interval of 25 years estimated and for Trans-Yamuna, Barapullah, Najafgarh basins (all having Catchment area more than 50 km^2) Unit hydrograph method was used for estimation of PMF. For both the approaches, contribution of the discharge from Average water supplied was also considered.

The flood of 25 years recurrence interval for Aruna Nagar and Old Chanderwal basin determined as a 5.12 and 19.93 cumecs. And Probable Maximum flood of Trans-Yamuna, Barapullah, Najafgarh basins computed as 761.81, 1996.74 and 4570.38cumecs.

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q_p =2.030/$(L/S^{0.5})^{0.649}$ …… Equation 6 ..... **19**

t_p=1.858/$(qp)^{1.038}$ …… Equation 7 ..... **19**

W_{50}=2.217/$(qp)^{0.990}$ …… Equation 8 ..... **19**

W_{75}=1.447/$(qp)^{0.876}$ …… Equation 9 ..... **20**

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Q_p=q_p*A …… Equation 13 ..... **20**

Q_n= $\sum^{n}_{i=1} P_i U_{n=i+1=}$ $P_n U_1$+ $P_{n-1} U_2$+……. +$P_1 U_n$……. Equation 14 ..... **20**

LIST OF ABBREVIATIONS

1. **PMP** Probable Maximum Precipitation
2. **PMF** Probable Maximum Flood
3. **UH** Unit Hydrograph
4. **MSL** Mean Sea Level
5. **GSDP** Gross State Domestic Product
6. **GSVA** Gross State Value Added
7. **DEM** Digital Elevation Model
8. **SRTM** Shuttle Radar Topography Mission
9. **NCT** National Capital Territory
10. **IIT** Indian Institute of Technology

CHAPTER 1
INTRODUCTION

1.1 Floods

When it comes to the effects of hydro-meteorological disasters like floods, storms, wet mass movements, severe temperatures, droughts, and wildfires, India is among the most significant nations in the world. Floods are one of the world's most devastating natural hazard that occur very frequently. They impact millions of people worldwide every year. It can cause loss of life, significant economic damages and long-term environmental consequences. The number of places at risk of flooding is increasing due to human activities including urbanization, deforestation, and uncontrolled development. Increased population density, unplanned urbanization, and lax government inspection of construction projects have all contributed to the enormous reoccurrence of floods. About 8 million hectares of India's land area are vulnerable to yearly flooding, making up around 12% (or 40 million hectares) of the country's total area that is susceptible to riverine and flash floods.

1.1.1 Types of floods

There are different kinds of floods which are categorized based on their causes and characteristics. These are:

- **Riverine Floods** arise when rivers overflow caused by ice jams, extreme precipitation, or snowmelt. River flooding can occur gradually over a few days or weeks, allowing towns to get ready.
- **Flash floods** are defined as severe, abrupt floods brought on by torrential rains or dam failures. These are very dangerous because they can happen in a matter of minutes or hours, frequently with little notice.
- **Coastal Floods**, storm surges, high tides, and sea level rise can cause flooding in coastal locations. Severe coastal flooding can result from the exacerbation of these circumstances by hurricanes and tropical storms.
- **Urban Floods**, happens when it rains a lot, urban areas with inadequate drainage systems may flood. Cities impervious surfaces keep water from penetrating the earth, causing it to collect in the streets and it goes in the building basements.
- **Pluvial Floods**, happen when a lot of rain exceeds the capacity of the local drainage system, resulting in water build-up in low-lying areas. Pluvial floods do not need to occur near a body of water, in contrast to river floods.

- **Glacial Lake Outburst Floods (GLOFs),** happen when a glacial lake-containing dam breaks and releases a significant volume of water downstream. GLOFs are typical in mountainous areas where glaciers are retreating.

The study is based on identifying reasons of urban floods in Delhi as well as estimating the maximum flood in various sub-basins of Delhi that can occur.

1.1.2 Urban Flood:

Urban flooding is the occurrence of land or property being inundated by severe rainfall that overwhelms drainage systems within a man-made environment, particularly in highly populated regions such as cities. Urban flooding is produced by excessive runoff in urbanized regions where water lacks suitable outlets, in contrast to natural floods caused by river overflow or storm surges. Furthermore, one of the main causes of the worsening flooding issue is the fluctuating weather, which has resulted an increase in the occurrence of extreme rainfall events. Urban floods are mostly caused when the inflow of water exceeds the river ability and the expansion of human settlements quickly without enhancing drainage infrastructure in floodplains and low-lying areas. This hazard is recurring, especially in metropolitan areas, and its effects on local, regional, and international economies make an effective flood modelling strategy imperative for mitigating it. Global urban population anticipated to rise from 2009's 3.4 billion to 2050's 6.3 billion. According to these figures, by 2050, almost 68% of the world's inhabitants is anticipated to be in cities. Many cities in India also faced floods over the years which points out the gaps in the drainage infrastructure of those cities. (Fig 1).

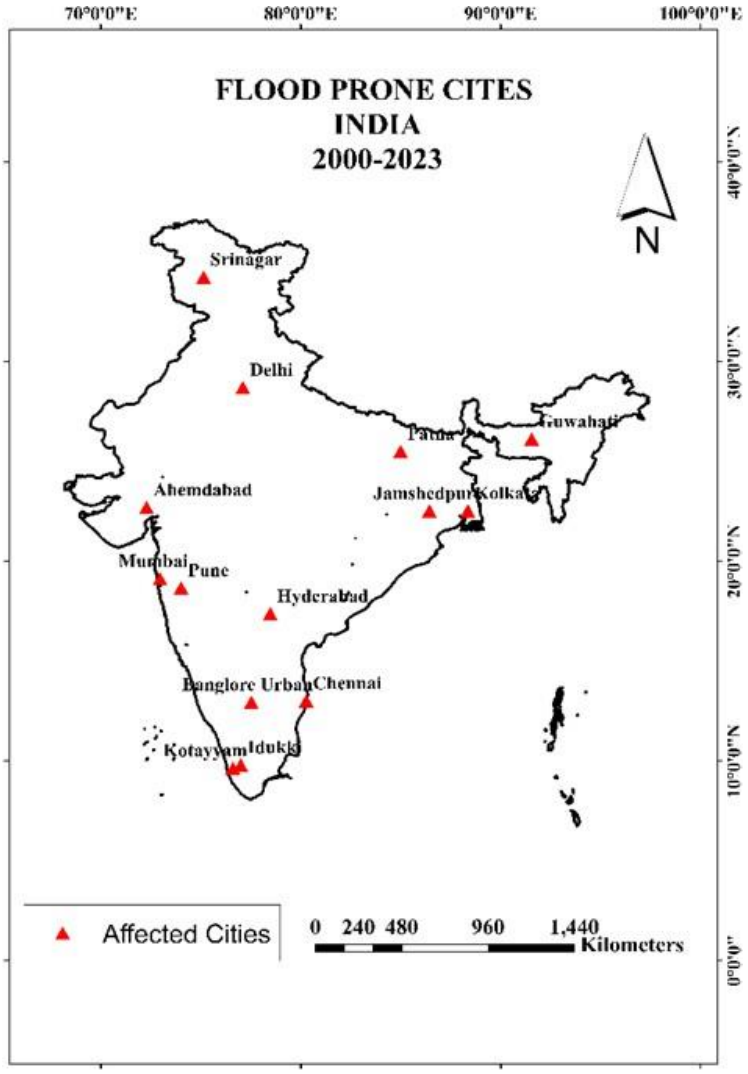

Figure 1 Map of Flood Prone Cities in India

1.1.3 Causes of Urban Floods:

- **Intense Rainfall:** Heavy rains are one of the main causes of urban floods. Rainfall that is both heavy and protracted can overload drainage systems, particularly in urban areas where significant amounts of water need to be handled fast. Strong rainfall is caused by several factors, including: -
    - **Climate Change:** Around the world, there's an upsurge in the occurrence and intensity of rainfall events due to climate change. Increased evaporation rates brought on by warmer temperatures might result in an increase in atmospheric moisture content and possibly stronger storms.
    - **Storm Systems:** Hurricanes, tropical storms, and other weather systems have the capacity to dump a lot of rain quickly, severely damaging metropolitan infrastructure.

- **Impermeable Surfaces:** Roads, walkways, parking lots, and buildings are examples of the impermeable surfaces that are highly concentrated in urban settings. Surface runoff is increased by these surfaces because they keep rainwater from penetrating the earth. High concentrations of impermeable surfaces increase the runoff factor due to which rainwater quickly overflows into streets and drainage systems when it is unable to seep into the earth, raising the possibility of floods.
- **Inadequate Drainage Systems:** Drainage system in many cities of the world is either outdated or insufficient for present requirements. Drains in those cities were built years ago and they do not withstand with the present level of human density or climate. Those drains were also not maintained regularly which adverse the situation and recurrence of flood in those regions can be easily seen. Clogging and not regular cleaning of drains resulting in overflowing of drains during heavy drains as they are carrying more than the designed capacity.

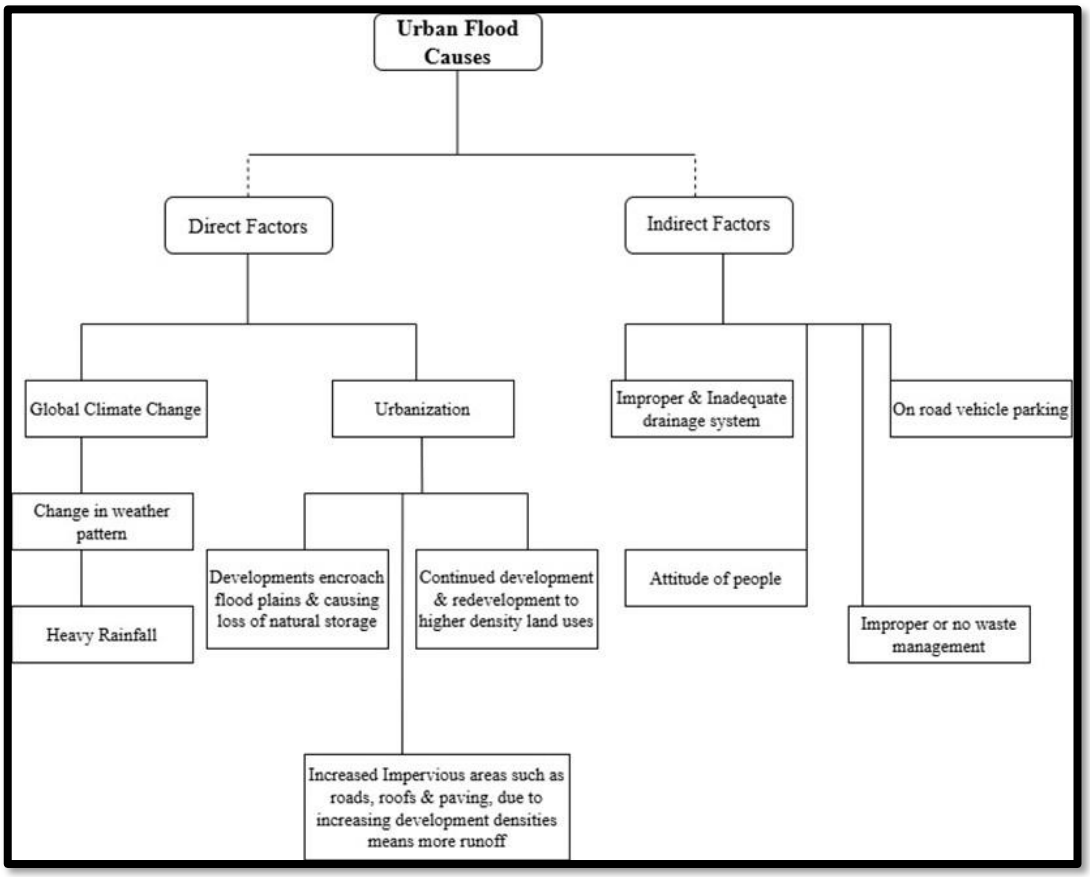

Figure 2: Causes of Urban Flood

1.1.4 Impacts of Urban Floods:

Urban flooding is an intricate issue which impacts the environment, infrastructure, society, economics, and urban development in several ways. These are:

- **Economic Effects:** Urban flooding can have detrimental effects on people's finances, as well as those of businesses and governments. Due to flooding major infrastructure gets damaged and the cost associated and repair is very high and a financial burden on impacted people and communities. Business also has to shut down all their operation during flood due to which the several people became jobless and the economy of the country is impacted. The recovery cost for emergency response by the government is also increased during that time and as well as for further future to prevent such flooding.
- **Social Impacts:** Urban flooding can have significant and far-reaching social effects on people, having an array of effects on both individuals and communities from displacement to education. During flood people may have to leave their homes permanently, which breaks them down emotionally as well as socially. Apart from this health risks are one of the major concerns as due to flood post-traumatic stress disorder **(PTSD)**, cholera, dysentery like diseases arises. Education of the children at the affected area also got affected and which pull them back from thriving towards success.
- **Environmental Impacts:** Due to urban flood all the waste materials of the urban area are carried by the floodwaters and end up in rivers, lakes etc. which pollutes the fresh water source of the area. Urban flood also causes soil erosion which results in loss of productive land as well as the sedimentation carried by the stream can be detrimental to the aquatic animals.
- **Impacts on Infrastructure and Transportation:** Flooding may severely damage infrastructure and transportation, which can result in road closures, which affects the everyday journeys and hinder emergency response activities. Further the public transit can get badly damaged which causes a lot of problems for the citizen. Regular recurrence of flood can cause roads bridges and buildings to lose their structural integrity, which further will deteriorate over and will cost way more than expected to maintain.
- **Impacts on Urban Resilience over Time:** Long-term effects of urban floods may include decreased resiliency and a reduced capacity for future crisis adaptation in cities, which enhance vulnerability due to experiencing flooding on a regular basis that may become less resilient to future floods and other climate-related problems. To increase

resilience and lower the risk of flooding, cities may need to make investments in adaptation and mitigation techniques, such as constructing flood defences, reclaiming natural floodplains, and enhancing drainage systems. Governments, corporations, and communities must work together to address urban floods, which should include innovative solutions that combine technology, natural methods, and community-based activities are also needed.

- **Impact on Urban Development:** Apart from all these regular urban floods also impacted the urban development as due to this land use of the area might changes which became a restriction for further infrastructure as well as for rezoning the area. Also, the drainage pattern of the newly planned area must be done in such a way that urban flood does not occur and this requires a lot of money. Adoption of any green infrastructure is very expensive and everyone cannot afford them. Building materials also changed and is also affect the cost of the infrastructure in the area.

1.2 MITIGATION & ADAPTATION STRATEGIES

Urban flooding mitigation solutions use the combination of policy changes, enhancements to infrastructure, and community involvement to reduce the potential and effects of floods. To develop resilient urban settings, this all-encompassing strategy combines engineering solutions with cutting-edge technologies and sustainable behaviours. Some of the mitigation strategies that can be adopted are:

- **Improving Storm water Drainage Systems:** Upgrading and maintaining storm water drainage systems is one of the best strategies to reduce urban flooding. To handle higher water volumes during periods of intense rainfall, storm water drains and sewers are being made larger and more capable in various cities around the world. Ensuring routine drain cleaning and inspection to avoid obstructions from trash, silt, and debris that can obstruct water flow which will reduce the recurrence of urban floods. In many cities of the world green infrastructure & blue infrastructure-based drains is been laid on which is known as advanced drainage solutions. In regions with combined sewer systems, keeping stormwater and sewage separate can help reduce contamination and flood risk by preventing overflows during periods of heavy precipitation.
- **Implementing Green Infrastructure:** Utilising landscape elements and natural processes to control stormwater and lower the danger of flooding is known as "green infrastructure." Planting plants on roofs to collect rainfall, lessen runoff, and improve insulation. Green roofs can create urban habitat and enhance the quality of the air.

Construction should be like in way which by which water can permeate the ground, porous materials can be used for roads, parking lots, and walkways. This lowers the risk of flooding and surface runoff. Creating planted spaces that collect and filter rainwater runoff so that it gradually seeps into the earth is known as rain gardens and bioswales. Both biodiversity and urban aesthetics can be improved by these elements.

- **Enhancing Land Use Planning & Zoning:** Improving Zoning and Land Use Planning which minimize the recurrence of urban floods in the cities. Restricting development in the regions which are vulnerable to floods especially the floodplains reduce the urban floods. Natural floodplains store the most amount of natural water flow as well as the excessive rainfall. Smart growth policies preserve open spaces and greenbelts that can absorb surplus water and offer recreational advantages while promoting high-density development in safe locations. To guarantee that new developments are not situated in high-risk areas and to guide land use planning, comprehensive evaluations of flood risk should be conducted.
- **Learning from Best practices and Case Studies:** Effective flood mitigation methods can be developed and implemented with the help of best practices and successful case studies. Prominent instances consist of:
    - **The Netherlands** is well-known for its creative approach to flood control. To reduce flood danger, the country combines engineering solutions, like barriers and dikes, with natural methods, like Room for the River. Implementation of the green infrastructure regular monitoring of the sewage and storm drains and the installation of the early warning system were the most prominent features of the tackling urban floods in the city.
    - **Singapore:** To lower the risk of flooding in a highly urbanised setting, Singapore has put in place a thorough drainage master plan that includes green infrastructure, infrastructural enhancements, and public education campaigns.
    - **New York City:** To improve urban resilience, the city created the Resilient New York plan in the wake of Hurricane Sandy. This plan involves investments in flood defences, climate adaptation, and community participation.

1.3 City Profile: DELHI

Delhi is National Capital of Republic of India & one of the major Urban cities in the nation. The city is located at an average elevation of about 220m or 720 feet above the mean sea level (MSL) and its geographical extent is from 28°37′N 77°14′E / 28.61°N 77.23°E. The state expands over 1483 sq. km of area in northern India sharing its border with the states of Haryana, Uttar Pradesh. It is near the edge of the Gangetic plains on the Yamuna River's right bank. It is primarily made up of alluvial plains with a lengthy rocky ridge that runs roughly from northeast to southwest. On the west bank of the Yamuna, the Aravalli Hills enter the southern border and terminate in the north of Delhi. One of India's most contaminated rivers is the Yamuna. It comes from the Yamunotri glaciers, which are around 6387 meters (about 3.97 mi) high in the lower Himalayas. One of the main causes of the increased river pollution in the Delhi NCT region is due to the settlement of people on its banks. River Yamuna has been divided into five segments. One of its Segment constitute as Delhi Segment which includes river barrages of Wazirabad to Okhla with an approximate length of segment measured around 22 km. It is home to around 16.78 million & about 97.5% people resides in urban areas **(Census, 2011)**. The city has experienced floods of varying magnitudes due to floods in the Yamuna and the Najafgarh Drain System. The Yamuna has exceeded its danger level (204.83m) 25 times in the last 33 years. Since 1900, Delhi has seen six significant floods: 1924, 1947, 1976, 1978, 1988, and 1995, with the top level of the Yamuna River being one meter or more beyond the danger level of 204.49m at the old rail bridge (2.66m above the danger level) on September 6, 1978. The second highest point, 206.92 meters, was reached on September 7, 1988. In recent years, the city has witnessed high magnitude floods in 1977, 1978, 1988, and 1995, bringing agony and loss of life. Over years the districts of Delhi most affected from flooding are the Central, North-East and East. The flood in June of 2023 was one of the devastating floods as the water level crosses the danger level and was flowing over 209m which can easily impact a lot of people by killing them and cause a huge damage to property. Due to Delhi's topography, rainwater travelled by the drainage system from the higher elevations in the west to the east, where it drained into the Yamuna. Naturally, the Yamuna flood plain included the low-lying Eastern side, which was deemed unusable because of the regular flooding.

Table 1 Water Level along with the Danger Level of various past floods of Delhi at Old Railway Bridge

YEAR	WATER LEVEL (m)	DANGER LEVEL
1976	**207.49**	**204.83**
1978	**207.49**	**204.83**
2013	**207.32**	**205.33**
2023	**208.48**	**205.33**

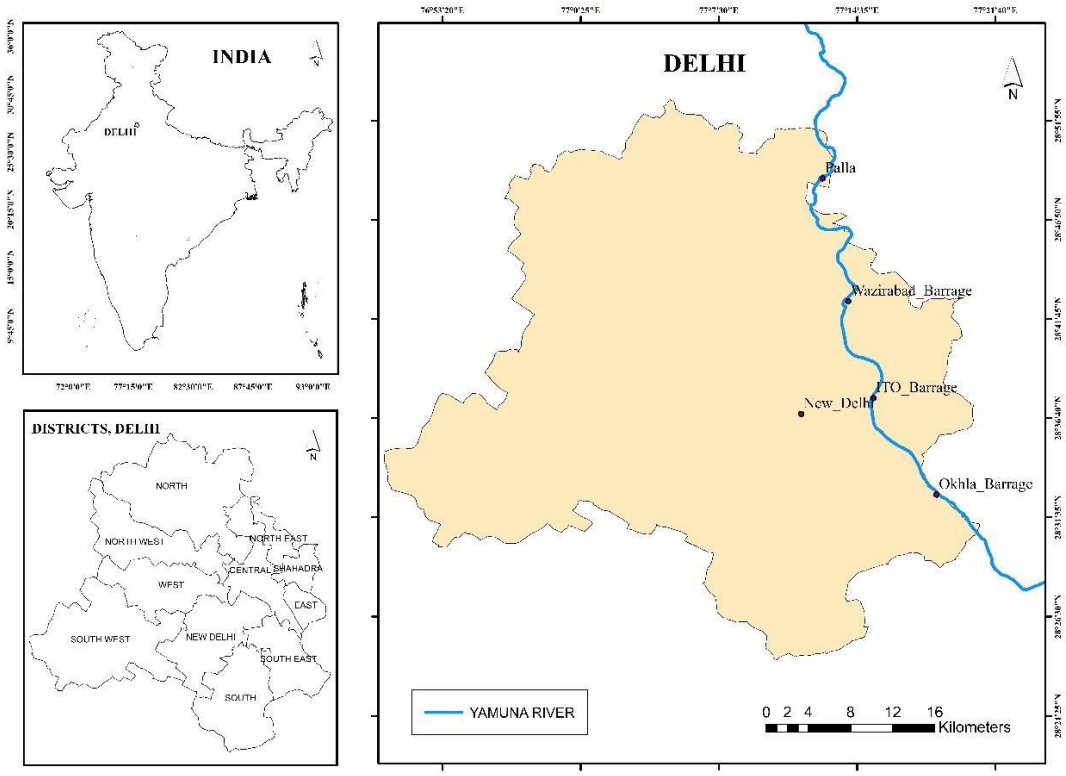

Figure 3: Map of Study Area

1.3.1 Drainage:

The Ganga basin's well-integrated drainage system includes the National Capital Region Delhi in general in the catchment number **403B**. The nearly uniformly mild gradient throughout the area limits the streams' and drains' ability to degrade. The area that makes up Delhi's metropolitan area is almost entirely made up of plain land that is distinguished by the existence of two major features: a river that enters the territory from the northeast and flows across its southeast and a long, rocky ridge that runs roughly from south to east. The Ridge serves as the local watershed, splitting the drainage system of the area into two sections. The ground in Delhi slopes generally from north to south. The western region uses the Najafgarh drain to flow into the Yamuna, whereas the eastern region empties directly into it.

The five main drainage basins of the National Capital Territory of Delhi are the **Old Chanderwal, Barapullah, Aruna Nagar, Najafgarh and Trans-Yamuna**. These divisions

are based on the basins made by Irrigation and Flood control Dep, Delhi. Because of the design of this drainage system, all water gathered in the major drains, link drains, and tiny rivulets is eventually released, either directly or indirectly, into the Yamuna. (DMP, 2018)

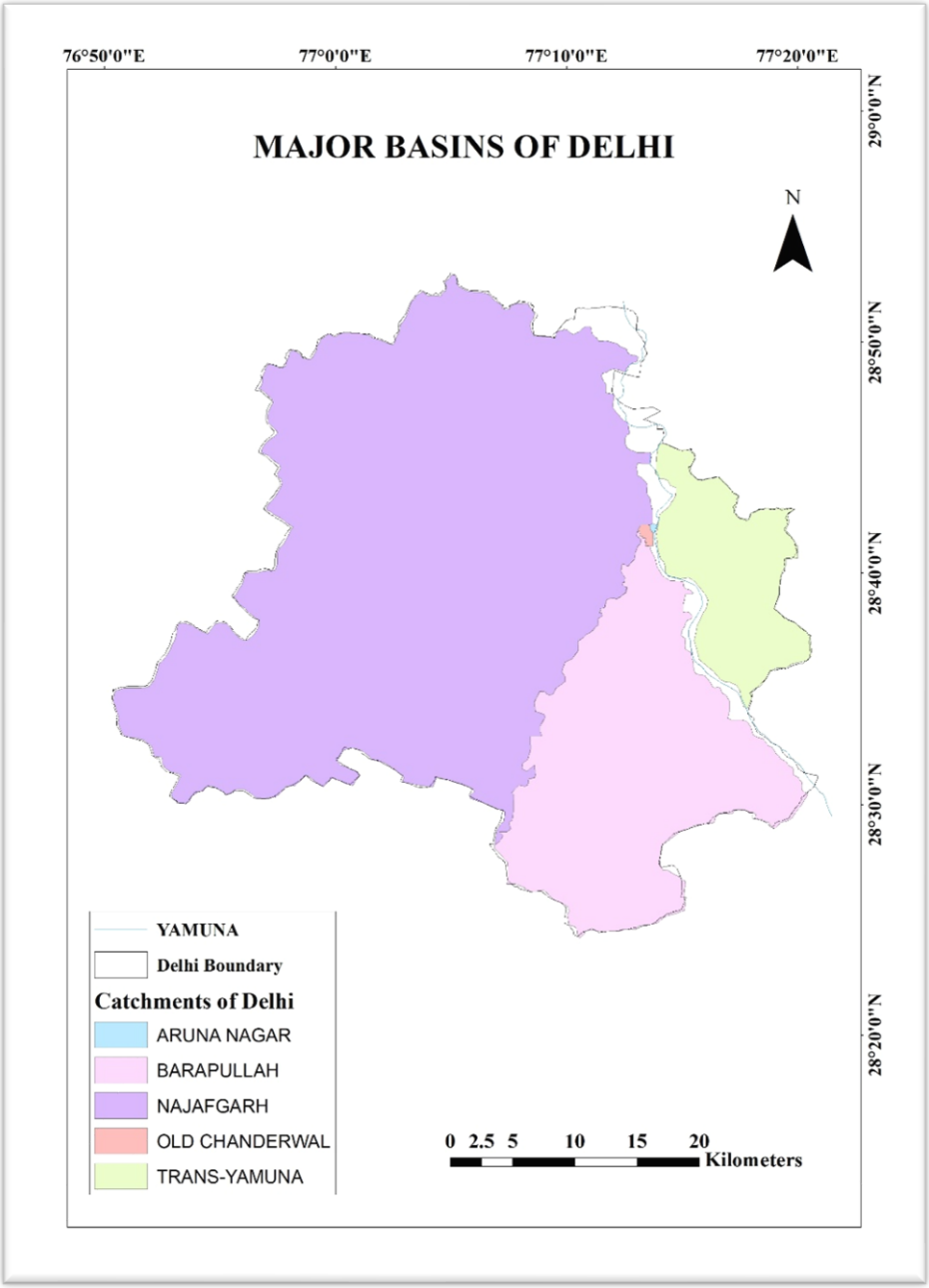

Figure 4: Major basins of Delhi

1.3.2 Geology:

Physio-graphically, Delhi is split into 4 sections: the ridge area, the newer alluvium east of the ridge, the ancient alluvium west of the ridge, and the Yamuna Ridge. It is in the Yamuna basin, which is composed of alluvium soils that were produced lately. Soil is composed of cankers, silt, gravel, clay, and fine to medium-sized sand. Alluvium sediments are composed of hard

formations. The Delhi system of rocks (Algonkian), which is composed of Alwar quartzite, is the oldest known formation. Alluvium is the most recent to sub-recent formation. The geological deposit that is intrusive after Delhi is basic and pegmatic. The city is surrounded by an established landscape with isolated, dispersed hills.

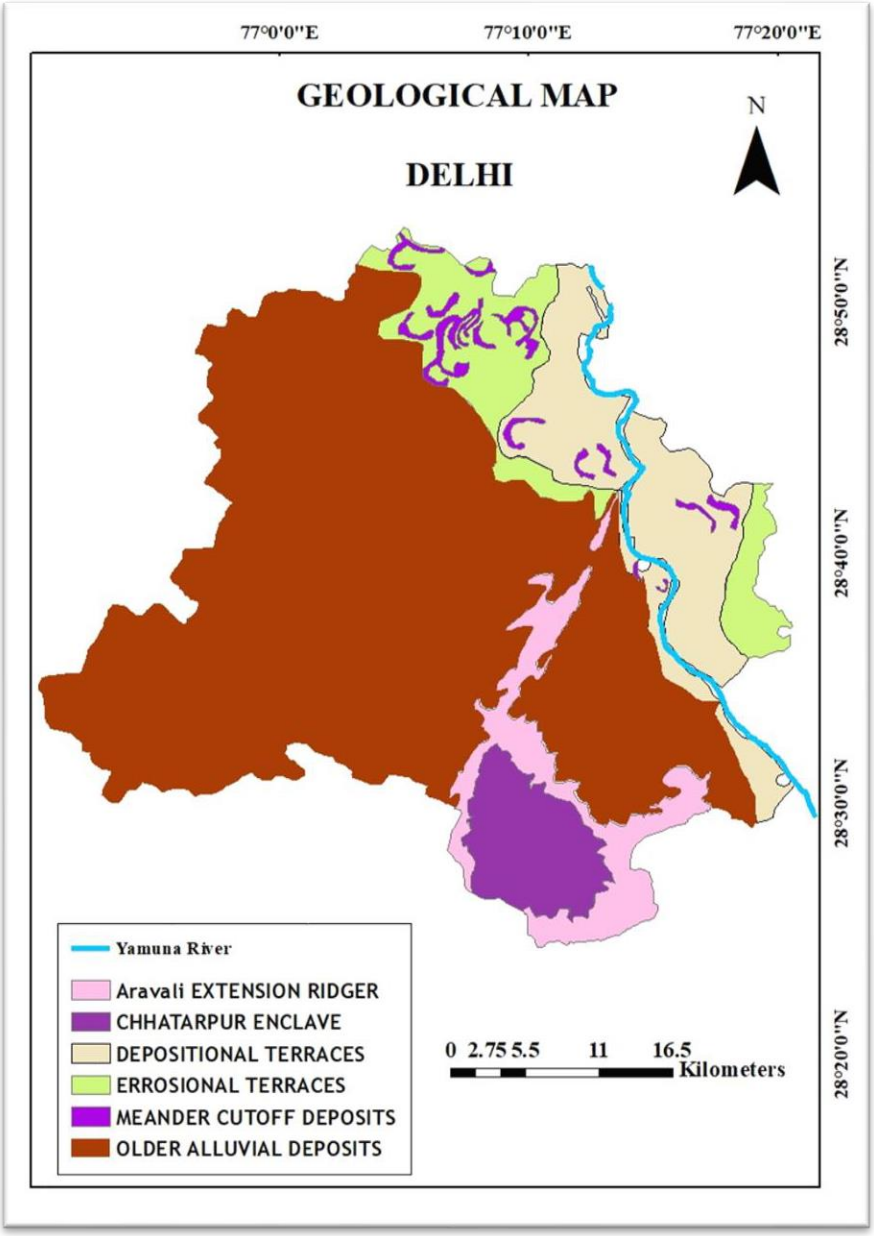

Figure 5: Geological & Soil map of Delhi

1.3.3 Demographic and Economic Profile:

Delhi ranks 13th in terms of Gross State Domestic Product (GSDP) among all the states and Union Territories. The state witnessed growth rate of 9.18% against a national average of 7% in the fiscal year 2022-2023. The per capita income is ₹ 444768, 2.6 times higher than the

national average. Delhi's economy has shown the same general trend of a larger participation in the service sector as most other urban economies. Over the years, Delhi's Gross State Value Added (GSVA) at current prices has shown a declining trend in the agricultural and allied sectors. The primary sector's percentage contribution to Delhi's GSVA at current prices has decreased from 3.49 percent in 2011–12 to 2.63 percent in 2022–23. The tertiary (service) sector's percentage contribution to Delhi's GSVA at current prices climbed from 83.42 percent in 2011–12 to 84.84 percent in 2022–23, while the secondary sector's contribution to the city's GSVA at current prices fell from 13.09 percent to 12.53 percent over the same period.

The city's resources, such as land, water, materials, and energy, are being stressed by the expanding population, but these needs cannot be adequately addressed. Usually, this results from a lack of appropriate and long-term management techniques. According to Hoyt et al. (2005), the lowest members of society are compelled to construct and occupy slums and squatter communities. The population of Delhi is made up of more than 45% slums, illegal colonies, and other unplanned settlements. (Gosain et al., ,2018) (DMP, IIT DELHI)

1.4 AIMS & OBJECTIVES:

- Estimation of Flood in various sub-basins using different approaches.
  - Rational Method
  - Unit Hydrograph
- Probable Maximum Flood from Critical Sequencing using Two Bells Approach.

CHAPTER 2
LITERATURE REVIEW

The study investigates the urban flood in Mumbai city that occur on 26 July 2005. It describes the drainage system of the city; the specifics of the flooding and the steps the city is doing to lessen similar disasters in the future. The post measures were forming a Chitale Committee (2006) it works with the Natu Committee (1974), CWRPS (1978), BRIMSTOWAD (1993) & IIT Bombay (2005) to identify causes of floods as well to suggest actions aimed at the future. The Chitale Committee restated that an insufficient drainage infrastructure was the primary cause of the disaster. The committee also suggested that the IMD install automatic rain gauges, measure, and create comprehensive contour maps of each watershed. It also suggested that the major thoroughfares and important municipal structures be subjected to 100 mm of rainfall per hour in the BRIMSTOWAD study. (Kumar Gupta, 2009)

The paper highlights the impact of unplanned urbanization and concretization of land surface as major factors responsible for floods in the city. The study also emphasizes solid waste and catchment management practices to reduce contamination in drainage system to mitigate floods. (Ramchandra T.V. & Mujumdar P., 2009)

The paper throw light on the unprecedented flood that occur in Hyderabad City in the year 2000. This report points out that the Fast industrialisation and urbanisation have resulted in unanticipated changes in land use, a surge in rural-urban migration, mounting strain on public infrastructure, and environmental deterioration. The natural drainage area for floods and rainfall has been reduced by construction, landscaping, and encroachment onto riverbeds surrounding the lake. Markandeya.K & Suryanarayana. G (2009)

The paper highlights the problem of population explosion in cities increases the vulnerability of urban landscapes to floods and flash floods. The study discusses numerous reasons for floods in major for cities of India: Delhi, Mumbai, Kolkata, and Chennai and a critical analysis of extreme rainfall in for mega cities. The author also suggests preventive measures and emphasizes that besides improving disaster management procedure and drainage redevelopment government must take measures to diversify industries and employment opportunities to prevent large scale migration and reduce the burden on limited resources available in the cities. (Singh G.P. et al, 2009)

The report stated assessment done using the rainfall data to analyses floods in Kerala in 2018 through computations of volume of run off generated by river basins and sub-basins in the state. The study suggested reviewing the storage capacity of reservoirs, widening the barrage for

efficient water discharge and creation of suitable storage reservoirs for mitigating floods. (Directorate Central Water Commission, 2018)

Flood is an exceptionally peak in a river, usually reached when the water surpasses its banks and submerges the surrounding area. It is commonly recognised that floods can result in significant harm to people and property, as well as economic losses owing to the disruption of normal business operations. For hydrologic design, the hydrograph of extreme floods and stages that correlate to flood peaks give useful information. Other techniques that can be used to determine the size of a flood peak include the following: (Upper-Indo-Ganga-Plains-Subzone-1e, 1973)

A. Rational Method
B. The Empirical Approach
C. Hydrograph Unit Method
D. Flood Frequency Analysis

A. Rational Method

The rational formula is only applicable to catchments that are smaller than 50 square kilometres. It takes account of rainfall that is uniformly intense and occurs over a very long period in the basin.

B. Empirical Formulae

The empirical formulae utilised to estimate the flood peak are mainly regional formulas that rely on significant catchment features and statistical correlations between the observed peaks. Most formulas ignore the flood frequency as a parameter and almost always use the catchment area as one that affects the flood peak.

C. Unit Hydrograph Method

Unit hydrograph technique can be applied to predict the peak flood hydrograph if the rainfall causing the flood, the catchment's infiltration characteristics, and the relevant unit hydrograph are available. Extreme rainfall conditions are utilised to create design storms, or the hydrograph of the excess rainfall that causes extreme floods, for design reasons.

D. Flood Frequency Analysis

Floods and other hydrology processes are incredibly complicated natural occurrences. They are exceedingly hard to model analytically since they are composites of several component characteristics. In addition to the method described above, there is another approach that can be used to estimate floods i.e. statistical frequency analysis.

The study computes estimation of flood response, contribution of flooding potential, percentage of flood volume for 25 sub-watersheds, the paper considers Snyder Unit Hydrograph (UG) approach with GIS based spatial database for calculating discharge at Lower Tapi Basin. (Anupam KS, 2015)

The paper highlights the estimation and use of IDF curves to rely on the hypothesis of rainfall series stationarity, namely that intensities and frequencies of extreme hydrological events remain unchanged over time. (Suchithra & Agarwal, 2020)

The study describes estimation of flood peak at the Hatnikund and Okhla Barrage sites in Yamuna River basin using frequency analysis and empirical formulae methods. Design floods for different return periods are computed using Gumbel's and Log-Pearson method. (Husain,2015)

The study describes flood peak estimation using two frequency analysis techniques: the Log-Pearson type III distribution and Gumbel's method. Different return-level flood magnitudes have been estimated using the daily discharge data. (Kumar et al., 2020).When the flood magnitudes derived using the POT technique and the annual maxima (AM) approach are compared, it is evident that the log-Pearson method yields generally higher flood magnitudes than Gumbel's method.

PMP is the amount of precipitation for a specific length of time that can be achieved methodologically for a given size storm area or design watershed at a specific place during a specific time of year. PMF is the highest flood that might be anticipated in the drainage basin due to the most extreme confluence of essential meteorological and hydrologic variables that are conceivably achievable. (Guidelines for Selecting and Accommodating Inflow Design Floods for Dams Guidelines for Selecting and Accommodating the Inflow Design Floods for Dams Central Water Commission, Ministry of Jal Shakti Department of Water Resources, River Development & Ganga Rejuvenation Government of India, 2021),

The report envisioning a future in which urban areas are generally small and medium-sized cities, are secure, inclusive, and well-off places to live and work that are resilient to a various threat. (Nations Development Programme, 2021)

The recent publication on *'Rapid Urban Growth in Flood Zones'* recognizes the disproportionate development in previously avoided areas such as floodplains, riverbeds and wetlands. The study shows that settlement growth in flood-prone areas is higher than safe areas and emphasizes on investments in disaster preparedness and protection and create urbanization plans to reduce expansion of cities in flood-prone areas to enable resilient socio-economic development. (Rentschler et al., 2022)

The handbook discussed the problem of urban flooding, hydrological and hydraulic approach to flood hazard assessment and suggests structural and non-structural measures to reduce the impact of urban flooding in the cites. (Assessing Risk and Identifying Interventions Urban Flood Risk Hand Book, 2023)

The study identifies urban heat island effect as the contributing factor in higher rainfall in urban areas over the years which causes the regular recurrence of floods in various cities. The study aims to establish rainfall measuring stations locally and densely to provide early flood warning. The guidelines mention the extensive guidelines prepared by urban development authorities for sewer drainage and a small section of drainage for storm water and its ineffective implementation in cities. Pre-monsoon desilting, removal of solid waste and removal of sediment are key measures which must be taken to ensure effective storm water drainage. City road levels and drain inlet connectivity must also be taken into consideration at the time of construction. Rainwater harvesting, creation of rain gardens, detention ponds and lined channels were other recommendations made to reduce the impact of flood. (National Disaster Management Guidelines Management of Urban Flooding, n.d.)

The report provides guidelines to states and UTs, the Model Building Bye Laws (MBBL), 2016 that the MoHUA (Ministry of Housing and Urban Affairs) developed for the States and UTs require the full rainwater harvesting plan to be included for plot sizes of 100 square meters or larger. The MoHUA published Standard Operating Procedures (SOP) for Urban Flooding in 2017, outlining the precise steps that the State Government, District Administration, and Urban Local Bodies must take (Urban and Regional Development Plans Formulation and Implementation (URDPFI) Guidelines. It offers guidance on sustainable storm water drainage system design, planning, and management along with preparedness strategies for flood response in urban areas, MoHUA issued a Manual on Storm Water Drainage Systems, 2019. One of the key components of Atal Mission for Rejuvenation and Urban Transformation (AMRUT) is storm water drainage. Storm Water Drainage has received ₹2,969 crore (4%) of the ₹77,640 crore total allotted for projects under AMRUT. Rejuvenation of water bodies and wells is one of the primary elements of the AMRUT 2.0 Scheme. Harvesting rainwater through storm water drains into a body of water that is not receiving sewage or effluent is one of the acceptable parts under this.

The report highlights interference of unplanned development with natural streams and watercourses, leading rivers to expand and grow. India's total area under urban settlements (7933 towns) grew from 77370.50 km^2 in 2001 to 102220.16 km^2 in 2011, representing

24850.00 km^2 of new land brought under urban usage. Every city should have flood mitigation plans (floodplain, river basin, surface water, etc.) that are fundamentally embedded in the master planning and general land use policy of the city to reduce the risk of urban flooding. (Standard Operating Procedure Urban Flooding, 2017) Although the management of urban floods is a new topic, it requires an interdisciplinary, integrative approach. To create effective, dependable, and most representative urban flood/disaster management systems, several factors must be considered. The following three stages of disaster management are covered by the Standard Operating Procedures (SOP) for an efficient and successful response to urban flooding.

The study throws light on the city's dependence on natural drainage and its poor condition due to dumping of garbage and other waste material. The study delineated flood management plan including delineating boundaries of natural channels and providing pumping stations at different locations to relieve drainage congestion. The study also emphasized on taking several administrative measures to improve flood management of the city. The organisation also created a master plan for Guwahati Metropolitan area-2025 aimed to improve socio-economic, transport, physical and social infrastructure and provide housing to expanding population. Review of Studies on Urban Floods in Guwahati from Flood knowledge to Urban Action, 2014,

The study carried out in various stages, in which we first done research and develop our understanding about the different concepts related to floods and its estimation. Then we collect the different data that is required for us to complete the study. The list of the data that we used is mentioned below in the table. Flow Diagram of how different stages is done is also mentioned in the next page.

3.1 Basic Data

- DEM data has been extracted from the SRTM having 30m resolution and from that we calculate the slope of the area and different sub-basins and soil map was also generated to look the distribution of the different soils over Delhi. Further the data for Land use/ Land cover taken from ESRI.
- Base for demarcation of sub-basins of Delhi has been taken from Irrigation & Flood Control (Delhi).
- Hourly Rainfall data for last 23 years was obtained from DSP, Pune of IMD for Palam rain Guage Station.
- Demographic data especially Population along with Average Supply to the individual household has been referred from "report on Drainage Master Plan, Delhi, 2018" and Central Public Health & Environmental Engineering Organisation (CPHEEO) manual of Ministry of Housing & Urban Affairs respectively.

Table 2: Various Organization from where Data is been procured

ORGANIZATION	DATA EXTRACTED
Sentinel 1-A Satellite	DEM data
IN&FC	Drainage Map
Geological Map (Soil Map) as per IIT Delhi report	Soil Map
ESRI	Land use/Land cover
IMD	PMP atlas

3.2 Peak Discharges from the Basins:

3.2.1 Using Rational Formula

For quantification of peak discharges from the basin of area < 50 sq.km, Rational Formula including runoff coefficient as 0.8 for urban area & intensity of rainfall for 25 years of recurrence interval have been used.

Q_p = C*i*A for t >=tc ..... Equation 1

Where,
Q_p= Peak Discharge

C= coefficient of runoff (runoff/rainfall)

t_c, time of concentration (minutes)

i= intensity of rainfall (mm/hr)

A= area of catchment (km^2)

Time of Concentration

The time taken for runoff to travel from the hydraulically most distant point of the watershed to a point of interest within a watershed.

Calculation for Time of concentration

$t_c = 0.01947\ L^{0.77}\ S^{-0.385}$ ……Equation 2

Where,

t_c= time of concentration (minutes)
L= maximum length of travel of water (m)
S= slope of the catchment

Calculation for Rainfall Intensity

$i_{tc,p} = Kt^x / (t_c+a)^n$ …….Equation 3

Where,

$it_{c,p}$= Rainfall intensity in mm/hr
k, x, a and n= Coefficients, values depend on area.
t= 25 years
t_c= time concentration

Calculation for Average Water Supply Contribution in the storm
Qs= 1.8*0.8*Average water supply per capita/day* population of the basin ..... **Equation 4**

3.3 Determination of Synthetic Unit Hydrograph:

This study covers sub-basins of Delhi, which falls in the category of Upper Indo Gangetic Plains of Subzone(1e), thereafter formulae for Unit hydrograph were used for sub-basins having area >50 sq.km (i.e. Barapullah, Najafgarh & Trans Yamuna)

Calculation for UH:

$L/S^{.5}$ ..... Equation 5

$q_p = 2.030/(L/S^{0.5})^{0.649}$ ..... Equation 6

$t_p = 1.858/(qp)^{1.038}$ ..... Equation 7

$W_{50} = 2.217/(qp)^{0.990}$ ..... Equation 8

$W_{75}=1.447/(qp)^{0.876}$ …… Equation 9

$WR_{50}=0.812/(qp)^{0.907}$ …… Equation 10

$WR_{75}=.606/(qp)^{0.791}$ …… Equation 11

$T_B=7.744(tp)^{0.779}$ …… Equation 12

$Q_p=q_p*A$ …… Equation 13

Where,

A= catchment in sq.km

L= Length of main stream along the river course in km

Lc = Length of main stream from a point opposite to centroid of the catchment area to the gauging site in km, along the main stream.

S= Slope in mm/km

T_B= Base width of Unit Hydrograph in hours

Q_p= Peak discharge of unit hydrograph in cumecs

q_p= Peak discharge of unit hydrograph per unit area in cumecs per sq.km

T_p= time from the centre of unit rainfall duration to the peak of unit hydrograph in hours.

W_{50}=Width of the U.G. measured at 50% of maximum discharge ordinate (Q_p) in hours.

W_{75}=Width of the U.G. measured at 75% of maximum discharge ordinate (Q_p) in hours

WR_{50}= Width of the rising side of U.G. measured at 75% of maximum discharge ordinate (Q_p) in hours.

WR_{75}= Width of the rising side of U.G. measured at 75% of maximum discharge ordinate (Q_p) in hours.

3.4 Determination of Maximum Flood:

Temporal distribution of Probable maximum Precipitation for 12 hours was considered after applying clock hour correction & areal correction for probable maximum flood estimation. The below mention is the equation to calculate the convolution of the maximum flood.

$Q_n= \sum^{n}_{i=1} P_i U_{n=i+1} = P_n U_1 + P_{n-1} U_2 + \ldots\ldots + P_1 U_n$ ..... **Equation 14**

Fig 6 shows the stages of how the study progresses in different phases.

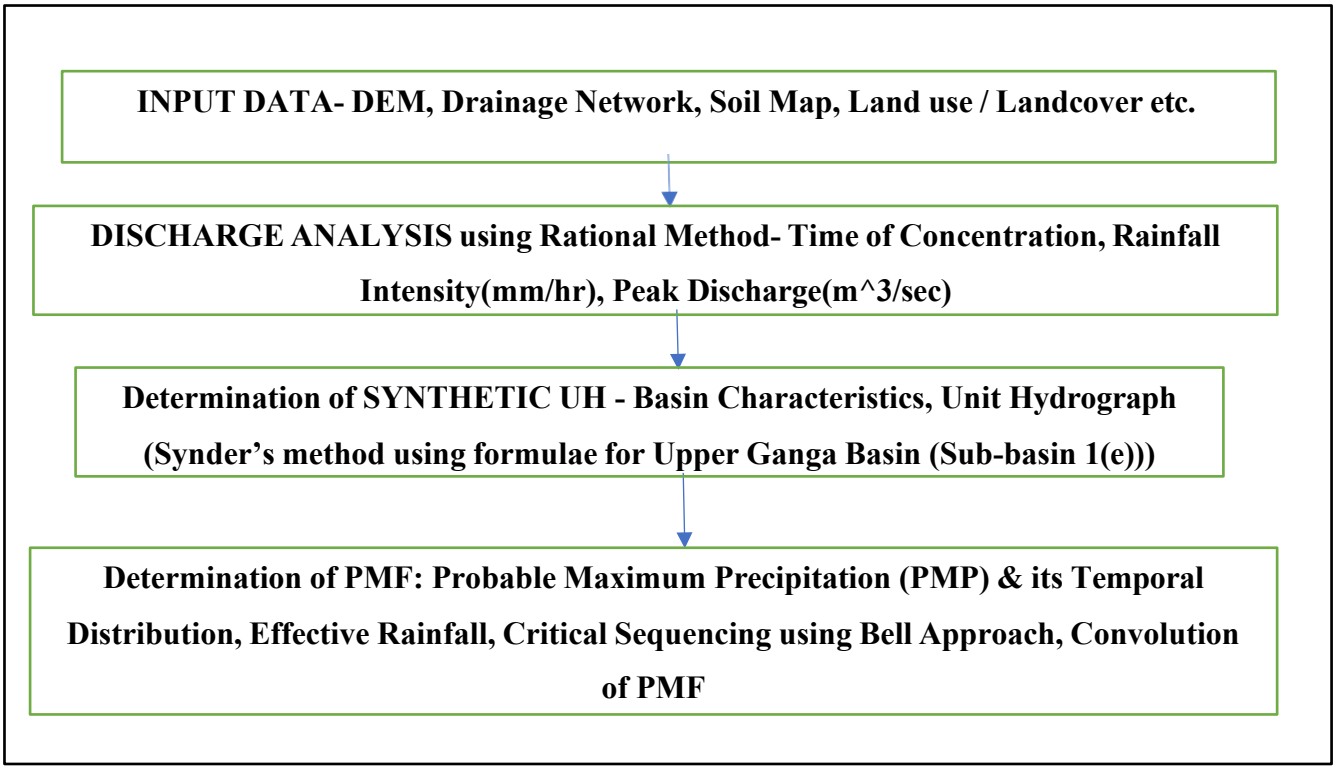

Figure 6: Methodology Adopted

This chapter includes the work done mainly in two phases. Initial phase covers determination of peak discharges occurring in various sub-basins of Delhi for 25 years of recurrence interval. For sub-basins of Delhi having area <50 sq.km, estimation of peak flood has been done using Rational Formulae. Thereafter in final stage, Sub-basins having area >50 sq.km estimation of probable maximum flood has been done based on Unit Hydrograph approach, since Probable Maximum Precipitation considered for the study. For both the approaches Contribution from average water supplied was also determined.

4.1 Peak Discharge Calculations using Rational Approach

Average Supply in each household has been considered as 200 LPCD for taking up the contribution of water supply, and contribution of water supplied for respective basin has been determined.

Table 3 Discharge Contribution from Water supply

Assumption: Avg Supply/Capita/day (LPCD)			**200.00**
S.N.	**Basin**	**Population (Lakhs)**	**Contribution from Avg. Supply (m^3/s)**
1	Aruna Nagar	300000	1.00
2	Old Chanderwal	300000	1.00
3	Trans Yamuna	3950000	13.17
4	Najafgarh	8738000	29.13
5	Barapullah	3389000	11.30

Maximum length of travel of water in a basin & longitudinal slope of basin determined using DEM data from SRTM having resolution of 30m interval with help of ArcGIS ver. 10.8. Time of concentration using Kirpich equation has been determined for calculating the rainfall intensity of 25 years of recurrence interval.

Table 4 Calculations for Max. Rainfall intensity for 25 years Recurrence interval for Sub-basins of Area<50 sq.km

S.N.	**Basin**	**Max. Length of Travel of Water (m), L**	**Longitudinal slope of the basin, S**	**Time of Concentration (hrs), $t_c=0.01947*L^{(0.77)}*S^{(-0.385)}$**	**Rainfall Intensity, $i_{c,p}$ (mm/h) = $5.91*(25)^{(0.1623)}/(t_c/60+0.5)^{(1.0127)}$**
1	Aruna Nagar	391	48.5	0.43	19.82
2	Old Chanderwal	916	49	0.83	19.56

Peak discharges for respective basins determined using rational method along with contribution of sewage water generated from average water supplied to individual households.

Table 5 Peak Discharge of sub-basins of Delhi NCT (Area<50 sq.km) using Rational Method

S.N.	Basin	Area (A) in Km^2	Runoff Coefficient, C	Rainfall Intensity, $i_{c,p}$ (mm/h)	Peak Discharge from Rainfall (m^3/s), Qp= C*i*A	Supplied Water Contribution, (m^3/s)	Total Peak Discharge, (m^3/s)
1	Aruna Nagar	0.26	0.80	19.82	4.12	1.00	5.12
2	Chanderwal	1.21	0.80	19.56	18.93	1.00	19.93

4.2 Peak Discharge Calculation using Unit Hydrograph (UH)

Determination of synthetic unit hydrograph parameters has been done using respective catchment characteristics like area, Length, Slope etc. (ref: CWC report on "**Indo Gangetic Plains of SubZone-1(e)"**).

4.2.1 Trans Yamuna Basin

Based on extracted maps of Trans Yamuna sub basin from ArcGIS we have taken the Area, L (in km), Lc (in km) and slope from which the parameters of Synthetic UH were calculated given in Table 6.

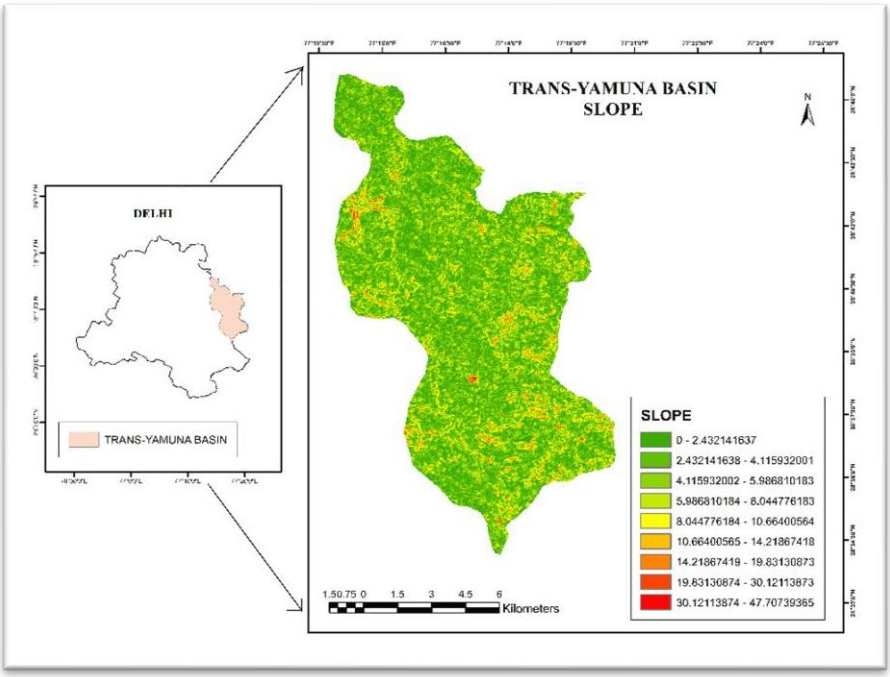

Figure 7 Catchment area Map of Trans Yamuna Basin

Table 6 Synthetic UH parameters for Trans Yamuna Sub Basin

Trans Yamuna		
Indo Gangetic Zone-1(e) CWC Report	**Calculated Values**	**Adopted Values**
A, sqkm	119.35	119.35
L, km	11	11
Lc, km	5.831	5.831
S	42.3	42.3
L/S^.5	1.69	1.69
qp=2.030/(L/S^0.5)^0.649	1.44	1.44
tp=1.858/(qp)^1.038	1.27	3.00
W50=2.217/(qp)^0.990	1.54	2.00
W75=1.447/(qp)^0.876	1.05	1.00
WR50=0.812/(qp)^.907	0.58	1.00
WR75=.606/(qp)^.791	0.45	1.00
TB=7.744(tp)^.779	10.31	10.00
Qp=qp*A	172.27	172.00

Unit Hydrograph Ordinates (in Table 7) for Trans Yamuna Basin determined using synthetic parameters as above in such a way that storm depth exactly matches as unity and Unit Hydrograph plot for Trans Yamuna Basin also determined as Fig 8.

Table 7 UH Ordinates for Trans Yamuna Sub-Basin

UH Ordinates	
Time (hrs)	**Runoff m^3/s**
0	0.0
1	30.0
2	90.0
3	172.0
4	50.0
5	28.0
6	20.0
7	5.0
8	2.0
9	1.0
10	0.0
Sum of UH Ordinates	454.0
Catchment Area	119.4
Storm Depth	1.0

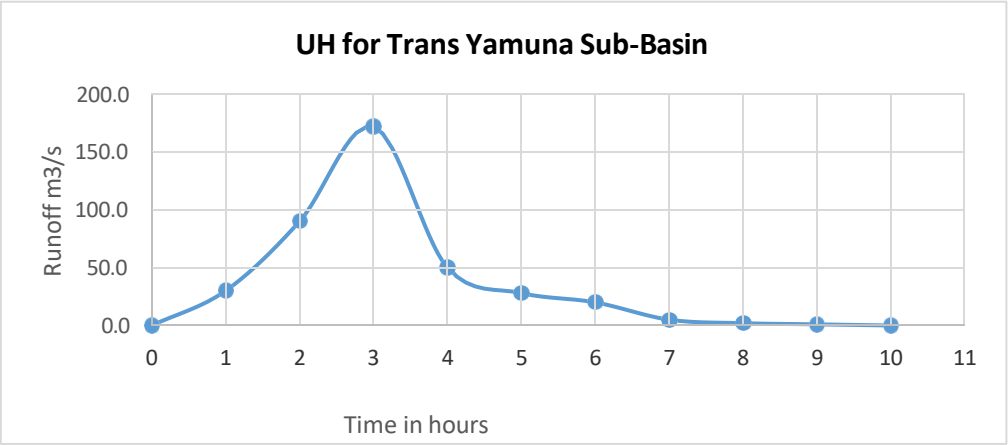

Figure 8 UH for Trans Yamuna Sub-Basin

4.2.2 Barapullah Basin

Based on extracted maps of Barapullah sub basin from ArcGIS we have taken the Area, L (in km), Lc (in km) and slope from which the parameters of Synthetic UH were calculated given in Table 8.

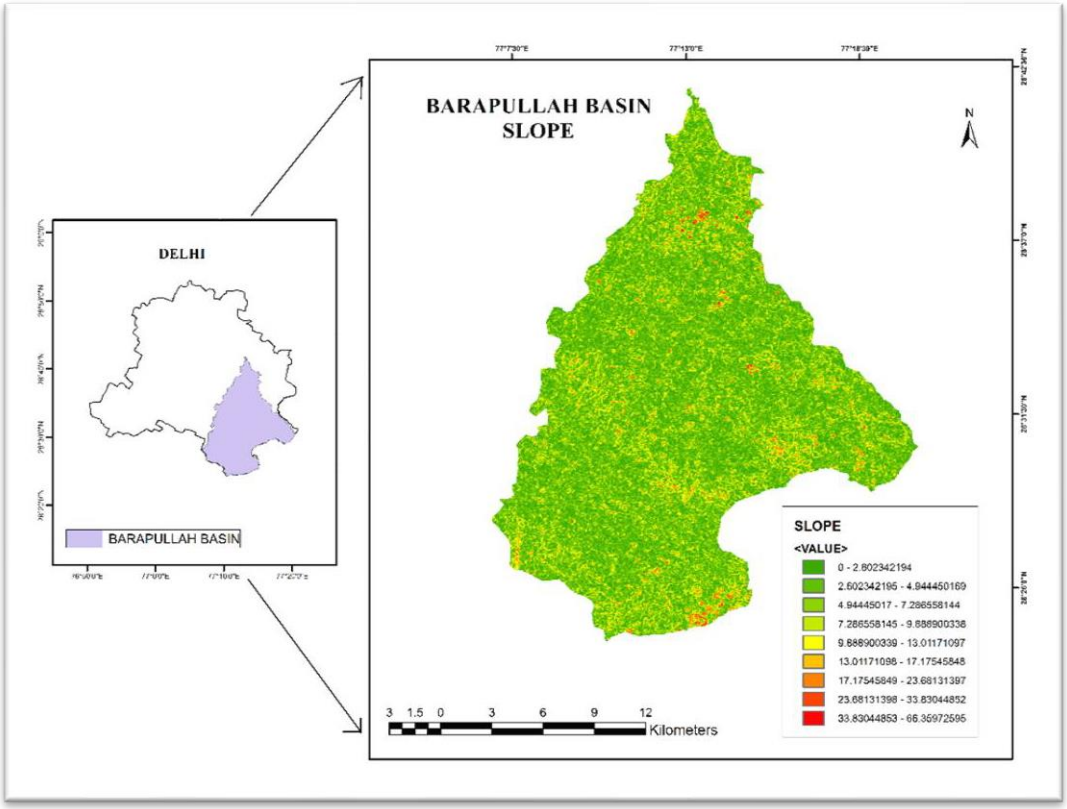

Figure 9 Catchment area Map of Barapullah Basin

Table 8 Synthetic UH parameters for Barapullah Basin

Barapullah		
Indo Gangetic Zone-1(e) CWC Report	**Calculated Values**	**Adopted Values**
A, sqkm	336.86	336.86
L, km	14.134	14.13
Lc, km	18.138	18.14
S	49.8	49.80
L/S^.5	2.00	2.00
qp=2.030/(L/S^0.5)^0.649	1.29	1.29
tp=1.858/(qp)^1.038	1.42	2.00
W50=2.217/(qp)^0.990	1.72	2.00
W75=1.447/(qp)^0.876	1.16	1.00
WR50=0.812/(qp)^.907	0.64	1.00
WR75=.606/(qp)^.791	0.49	1.00
TB=7.744(tp)^.779	9.46	10.00
Qp=qp*A	435.69	436.00

Unit Hydrograph Ordinates (in Table 9) for Barapullah Basin determined using synthetic parameters as above in such a way that storm depth exactly matches as unity and Unit Hydrograph plot for Barapullah Basin also determined as Fig 10.

Table 9 UH Ordinates for Barapullah Sub-basin

UH Ordinates	
Time (hrs)	**Runoff m^3/s**
0	0.0
1	200.0
2	436.0
3	228.0
4	100.0
5	50.0
6	20.0
7	5.0
8	2.0
9	1.0
10	0.0
Sum of UH Ordinates	1169.0
Catchment Area	336.56
Storm Depth	1.0

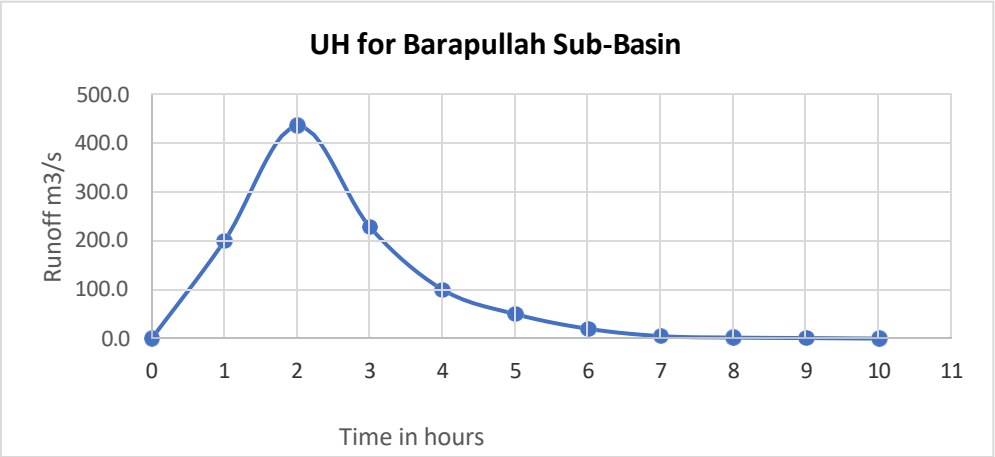

Figure 10 UH for Barapullah Sub-Basin

4.2.3 Najafgarh Basin

Based on extracted maps of Najafgarh sub basin from ArcGIS we have taken the Area, L (in km), Lc (in km) and slope from which the parameters of Synthetic UH were calculated given in Table 10.

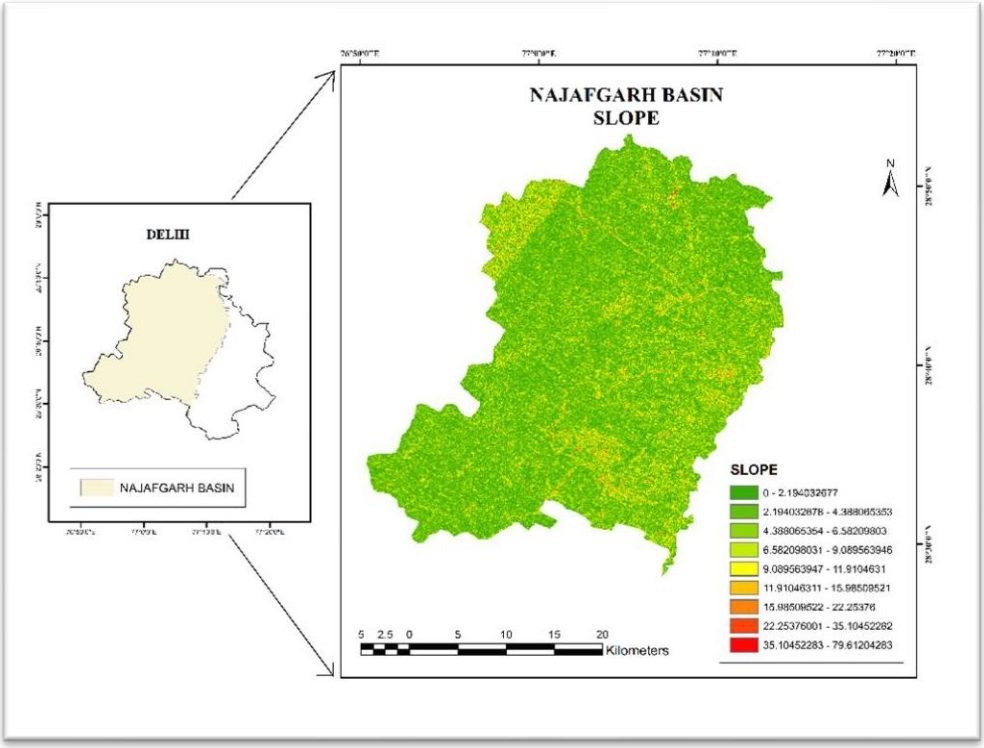

Figure 11 Catchment Area Map of Najafgarh Basin

Table 10 Synthetic UH parameters for Najafgarh Basin

Najafgarh		
Indo Gangetic Zone-1(e) CWC Report	**Calculated Values**	**Adopted Values**
A, sqkm	970.73	970.73
L, km	24.47	24.47
Lc, km	18.318	18.318
S	43.54	43.54
L/S^.5	3.71	3.71
qp=2.030/(L/S^0.5)^0.649	0.87	0.87
tp=1.858/(qp)^1.038	2.15	3.00
W50=2.217/(qp)^0.990	2.55	3.00
W75=1.447/(qp)^0.876	1.64	2.00
WR50=0.812/(qp)^.907	0.92	1.00
WR75=.606/(qp)^.791	0.68	1.00
TB=7.744(tp)^.779	6.93	7.00
Qp=qp*A	841.76	841.00

Unit Hydrograph Ordinates (in Table 11) for Barapullah Basin determined using synthetic parameters as above in such a way that storm depth exactly matches as unity and Unit Hydrograph plot for Najafgarh also determined as Fig 12

Table 11 Ordinates for Najafgarh Sub-basin

UH Ordinates	
Time (hrs)	Runoff m^3/s
0	0.0
1	500.0
2	750.0
3	841.0
4	745.0
5	200.0
6	50.0
7	0.0
Sum of UH Ordinates	3337.0
Catchment Area	970.73
Storm Depth	1.0

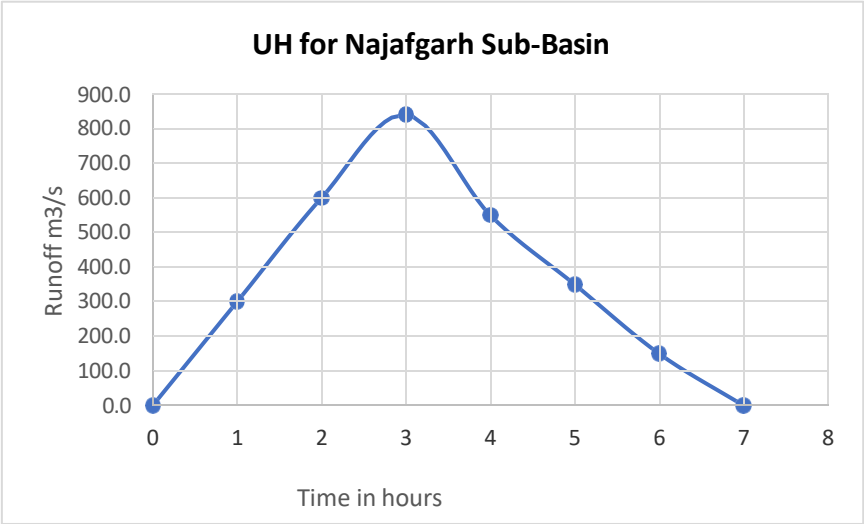

Figure 12 UH for Najafgarh Sub-Basin

4.3 Critical Sequencing

Base width of all UH for sub-basins is coming less than 12 hours therefore variation of daily rainfall for 12 hours has been considered. Table 12 illustrates the distribution coefficients of rainfall data for 12 hours.

Table 12 Temporal Distribution for all sub-basin for 12 hours rainfall

Time (Hours)	Distribution Coefficient
1	0.202
2	0.313
3	0.452
4	0.544
5	0.607
6	0.653
7	0.721
8	0.772
9	0.860
10	0.929
11	0.972
12	1.000

Effective rainfall using PMP after applying corrections for clock hour & areal extent has been determined in Table 13. Loss rate as 1 mm/hr has been adapted as infiltration loss. Two bells of 12 hours rainfall variation has been done since, 2 bell approach has been adopted for estimation of Probable maximum flood.

Table 13 Rainfall Excess for All Sub-basins (For Palam rain gauge station)

Time (Hours)	Distribution Coefficient	Contributing Depth (mm)	Depth of Rainfall (mm)	Loss (mm)	Effective Rainfall (mm)
For First Bell					
1	0.202	26.82	26.82	1.00	25.82
2	0.313	41.55	14.74	1.00	13.74
3	0.452	60.01	18.45	1.00	17.45
4	0.544	72.22	12.21	1.00	11.21
5	0.607	80.59	8.36	1.00	7.36
6	0.653	86.69	6.11	1.00	5.11
7	0.721	95.72	9.03	1.00	8.03
8	0.772	102.49	6.77	1.00	5.77
9	0.860	114.17	11.68	1.00	10.68
10	0.929	123.33	9.16	1.00	8.16
11	0.972	129.04	5.71	1.00	4.71
12	1.000	**132.76**	3.72	1.00	2.72
For Second Bell					
1	0.202	4.08	4.08	1.00	3.08
2	0.313	6.32	2.24	1.00	1.24
3	0.452	9.13	2.81	1.00	1.81
4	0.544	10.98	1.86	1.00	0.86
5	0.607	12.25	1.27	1.00	0.27
6	0.653	13.18	0.93	1.00	0.00
7	0.721	14.56	1.37	1.00	0.37
8	0.772	15.59	1.03	1.00	0.03
9	0.860	17.36	1.78	1.00	0.78
10	0.929	18.76	1.39	1.00	0.39
11	0.972	19.62	0.87	1.00	0.00
12	1.000	**20.19**	0.57	1.00	0.00
	Total		**152.95**		**129.59**

4.3.1 Trans Yamuna:

The above Rainfall excess ordinates are arranged in critical order for each bell separately. The largest depth is placed against the peak of Unit graph, the next largest against the next largest UH ordinate and so on until all rainfall increments get arranged. The rainfall sequence is then reversed to get the critical sequence as in Table 14.

Table 14 Critical Sequencing for Trans Yamuna

Time (Hours)	UH ordinates Flow	First bell - RF inc. (cm)	Critical Sequence (cm)	Time (Hours)	UH ordinates Flow	2nd bell - RF inc. (cm)	Critical Sequence (cm)
		Ist Bell				IIndBell	
1	0.00	0.58	**0.27**	13	0.00	0.308	0.000
2	30.00	1.12	0.47	14	30.00	0.086	0.000
3	90.00	1.75	0.51	15	90.00	0.181	0.000
4	172.00	2.58	0.74	16	172.00	0.308	0.003
5	50.00	1.37	0.80	17	50.00	0.124	0.027
6	28.00	1.07	0.82	18	28.00	0.078	0.037
7	20.00	0.82	1.07	19	20.00	0.037	0.078
8	5.00	0.80	1.37	20	5.00	0.027	0.124
9	2.00	0.74	2.58	21	2.00	0.003	0.308
10	1.00	0.51	1.75	22	1.00	0.000	0.181
11	0.00	0.47	1.12	23	0.00	0.000	0.086
12	0.00	**0.27**	0.58	24	0.00	0.000	0.308

Convolution for PMF for Trans Yamuna Basin

For obtaining the Probable Maximum Flood, UH ordinates are multiplied with effective rainfall for each hour. For taking up the contribution of effective rainfall for 2 hour the UH ordinated were lagged by 1 hour and so on. Further addition of all ordinated including contribution from average water supplied to the basin has been done. Maximum ordinate out from the total is assumed as PMF, Ref. **Annexure - A**. The UH convolution formula is:

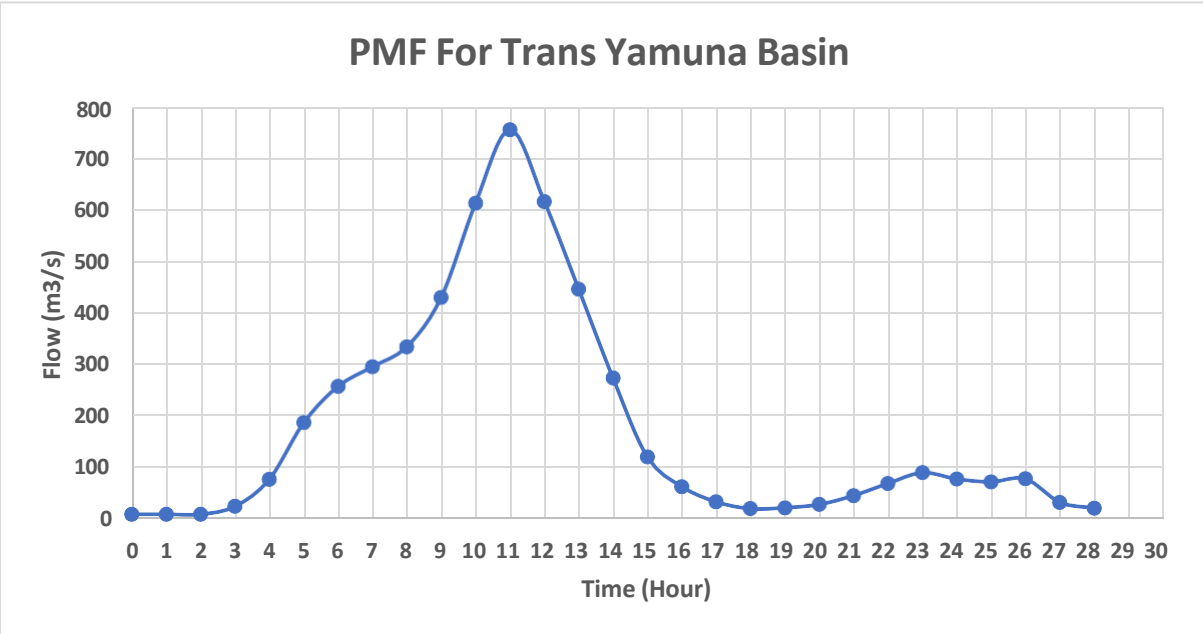

Figure 13 PMF for Trans Yamuna Sub Basin

4.3.2 Barapullah:

The above Rainfall excess ordinates in Table no 13 are arranged in critical order for each bell separately. The largest depth is placed against the peak of Unit graph, the next largest against the next largest UH ordinate and so on until all rainfall increments get arranged. The rainfall sequence is then reversed to get the critical sequence as in Table 16.

Table 15 Critical Sequencing for Barapullah Sub-basin

Time (Hours)	UH ordinates Flow	First bell - RF inc. (cm)	Critical Sequence (cm)	Time (Hours)	UH ordinates Flow	2nd bell - RF inc. (cm)	Critical Sequence (cm)
		Ist Bell				IInd Bell	
1	200.00	1.37	**0.27**	13	200.00	0.124	0.000
2	436.00	2.58	0.47	14	436.00	0.308	0.000
3	228.00	1.75	0.51	15	228.00	0.181	0.000
4	100.00	1.12	0.58	16	100.00	0.086	0.003
5	50.00	1.07	0.74	17	50.00	0.078	0.027
6	20.00	0.82	0.80	18	20.00	0.039	0.037
7	5.00	0.80	0.82	19	5.00	0.037	0.039
8	2.00	0.74	1.07	20	2.00	0.027	0.078
9	1.00	0.58	1.12	21	1.00	0.003	0.086
10	0.00	0.51	1.75	22	0.00	0.000	0.181
11	0.00	0.47	2.58	23	0.00	0.000	0.308
12	0.00	**0.27**	1.37	24	0.00	0.000	0.124

Convolution for PMF for Barapullah Basin

For obtaining the Probable Maximum Flood, UH ordinates are multiplied with effective rainfall for each hour. For taking up the contribution of effective rainfall for 2 hour the UH ordinated were lagged by 1 hour and so on. Further addition of all ordinated including contribution from average water supplied to the basin has been done. Maximum ordinate out from the total is assumed as PMF, Ref. **Annexure - B**.

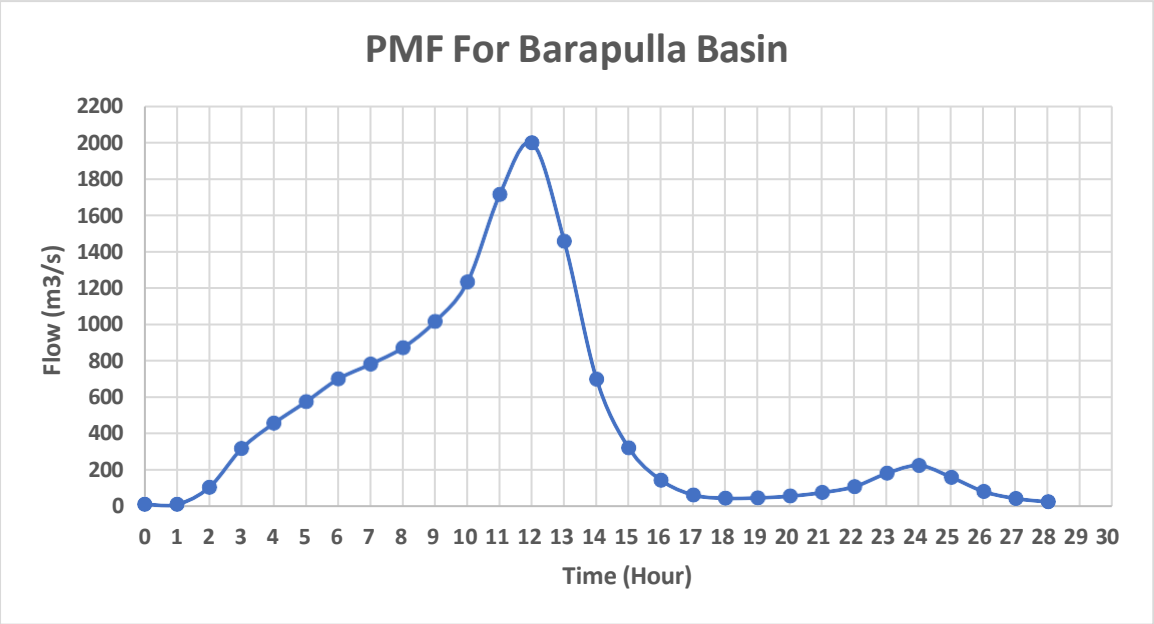

Figure 14 PMF for Barapullah Sub Basin

4.3.3 Barapullah:

The above Rainfall excess ordinates in Table no 13 are arranged in critical order for each bell separately. The largest depth is placed against the peak of Unit graph, the next largest against the next largest UH ordinate and so on until all rainfall increments get arranged. The rainfall sequence is then reversed to get the critical sequence as in Table 18.

Table 16 Critical Sequencing for Najafgarh Sub-basin

Time (Hours)	UH ordinates Flow	First bell - RF inc. (cm)	Critical Sequence (cm)	Time (Hours)	UH ordinates Flow	2nd bell - RF inc. (cm)	Critical Sequence (cm)
		Ist Bell				IInd Bell	
1	0.00	0.80	**0.27**	13	0.00	0.039	0.000
2	300.00	1.12	0.47	14	300.00	0.086	0.000
3	600.00	1.75	0.51	15	600.00	0.181	0.000
4	841.00	2.58	0.58	16	841.00	0.308	0.003
5	550.00	1.37	0.74	17	550.00	0.124	0.027
6	350.00	1.07	0.82	18	350.00	0.078	0.037
7	150.00	0.82	1.07	19	150.00	0.037	0.078
8	0.00	0.74	1.37	20	0.00	0.027	0.124
9	0.00	0.58	2.58	21	0.00	0.003	0.308
10	0.00	0.51	1.75	22	0.00	0.000	0.181
11	0.00	0.47	1.12	23	0.00	0.000	0.086
12	0.00	**0.27**	0.80	24	0.00	0.000	0.039

Convolution for PMF Hydrograph for Najafgarh Basin

For obtaining the Probable Maximum Flood, UH ordinates are multiplied with effective rainfall for each hour. For taking up the contribution of effective rainfall for 2 hour the UH ordinated were lagged by 1 hour and so on. Further addition of all ordinated including

contribution from average water supplied to the basin has been done. Maximum ordinate out from the total is assumed as PMF, Ref. **Annexure - C**.

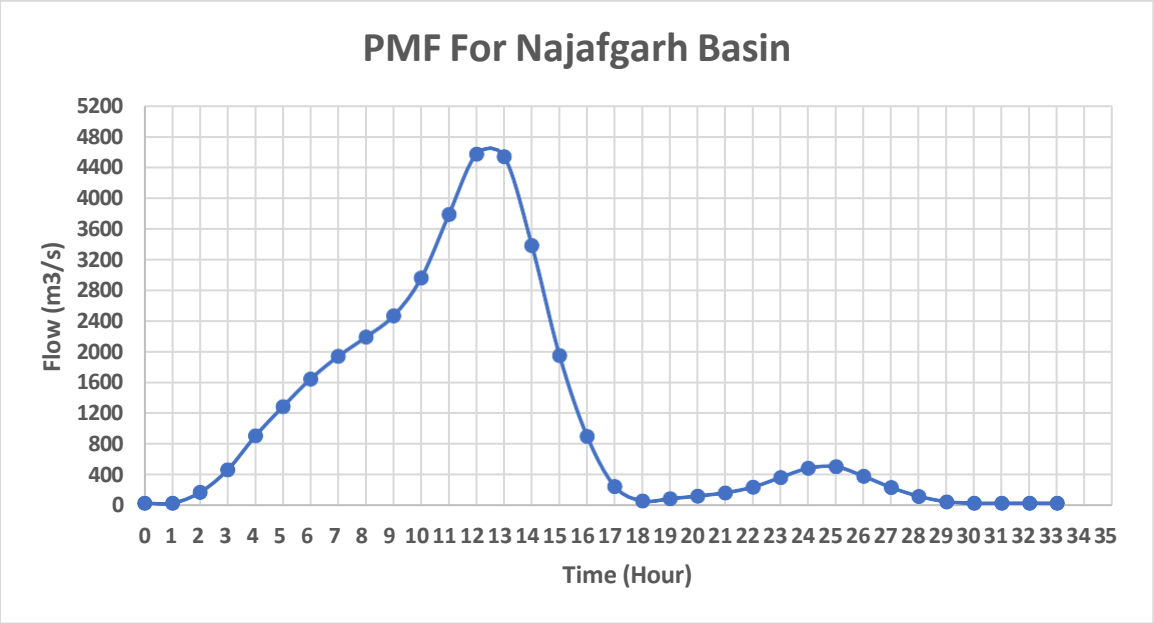

Figure 15 PMF for Barapullah Sub Basin

Design Flood

This study estimates the design flood using the maximum rainfall intensity corresponding to a 25-year return period recorded at the Palam rain gauge station.. For determination of Rainfall intensity, Time of concentration was determined first. Computed results for Design flood for various basins of Delhi NCT are tabulated in table 17.

Table 17 Design Flood of Various Sub-basins of Delhi NCT

Sr No.	Basin Name	Probable Maximum Flood (m^3/s)
1	Aruna Nagar	5.12
2	Old Chanderwal	19.93

The flood corresponding to a 25-year recurrence interval for the Aruna Nagar and Chanderwal basins was estimated as 5.12 and 19.93 cumecs respectively.

Probable Maximum Flood

For PMF estimation, the PMP of the Palam rain gauge station was used. Thereafter Unit hydrographs using various basin characteristics for each sub-basins (area> 50 sq-km) developed. Critical sequencing of Effective rainfall assuming losses as 1 mm/hr was carried outs, finally summing up the contributions of each effective rainfall spell using convolution was done.

Computed results for Design flood and PMF for various basins of Delhi NCT are tabulated in table 17 & 18 respectively.

Table 18 PMF of Various Sub-Basins of Delhi NCT

Sr No.	Basin Name	Probable Maximum Flood (m^3/s)
1	Trans Yamuna	761.81
2	Barapullah Basin	1996.74
3	Najafgarh Basin	4570.38

Probable Maximum flood of Trans-Yamuna, Barapullah, Najafgarh basins computed as 761.81, 1996.74 and 4570.38 cumecs.

Conclusions

- Rational methods estimate only the peak discharge and do not represent the variation of discharge across the basin.
- Unit hydrograph approach gives exact estimation of flood for larger basins (Area > 200 km^2) and it is accepted worldwide.

Gaps Identified

- The quantum of flood for each sub-basin requires clear passage of its flowing area to route without causing havoc to communities and infrastructure developed on the banks. Therefore, settlement in the nearby areas of drainage channels, streams should be engineered.
- Data dissemination of Rainfall and its variation over larger period plays a vital role on determination of floods.
- It has also been concluded that timely mapping of drainage network should be done.

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